

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	3-February-2026	Section	6A , 6B
	Exam Type:	Quiz 1 - Chapter 1	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[CLO 1] [5 marks]

- Which of the following best describes the primary disadvantage of strict protocol layering (e.g., the OSI or TCP/IP model)?
 - It prevents the abstraction of complex hardware details.
 - It increases the complexity of error detection at the physical layer.
 - It may lead to redundant functionality (e.g., error recovery) being implemented in multiple layers.
 - It makes it impossible to change the implementation of one layer without affecting the others.
- Time needed to perform an integrity check, lookup packet information in a local table and move the packet from an input link to an output link in a router.
 - Queueing delay
 - Processing delay
 - Transmission delay
 - Propagation delay
- A link has a high bandwidth-delay product. This implies:
 - The link is physically short but very fast.
 - A large volume of data can be in flight before the first bit is acknowledged.
 - The propagation delay is negligible.
 - Retransmissions are inexpensive.
- If the Traffic Intensity is close to 1, what is the expected behavior of the queueing delay?
 - It approaches zero.
 - It becomes constant.
 - It grows exponentially/asymptotically toward infinity.
 - It is equal to the propagation delay.
- A _____ is formed at the transport layer by encapsulating an application layer message with transport layer headers.
 - Segment
 - Datagram
 - Message
 - Frame

Q2: Suppose a caravan has 20 cars, and the tollbooth services a car at a rate of one car per 2 seconds. Once serviced, a car proceeds to the next toll booth, which is 400 kilometers away at a rate of 10 kilometers per second. The entire caravan must be queued at a tollbooth before the first car in the caravan can be serviced (i.e., pay its toll and begin driving towards the next tollbooth). Please answer the following questions: **[2+2+4+1+1 = 10 Marks] [CLO 1]**

Hint: This is an example of Store-and-Forward packet switching.

a. How long does it take for the entire caravan to receive service at the tollbooth (that is the time from when the first car enters service until the last car leaves the tollbooth)?

b. Once the first car leaves the tollbooth, how long does it take until it arrives at the next tollbooth?

c. Once the first car leaves the tollbooth, how long does it take until it enters service at the next tollbooth?

d. Are there ever two cars in service at the same time, one at the first tollbooth and one at the second tollbooth? And Why?

e. Are there ever zero cars in service at the same time, i.e., the caravan of cars has finished at the first tollbooth but not yet arrived at the second tollbooth? And Why?